1	2	3	$\sum$

Algorithmen, Datenstrukturen und Datenabstraktion (WiSe 18/19)

Docent: Prof. Dr. Wolfgang Mulzer

Tutorin: Valeria Zahoransky; Tutorial: 07

Exercise 09

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Viktoriya Kraleva, Samuel Enderwitz

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1 Expected Path Length in a Binary Search Tree (__/10)

Let n be a natural number. The randomly constructed binary search tree *with n nodes is obtained by inserting the keys from $\{1, \ldots, n\}$ in random order into a binary search tree that is initially empty (without rebalancing). Each of the $n!$ possible insertion orders is equally likely.*
Let T be a binary search tree and x a node in T. Let $d(x, T)$ denote the depth of x in T. The internal path length *$P(T)$ of T is defined as*

$$P(T) = \sum_{x \in T} d(x, T).$$

We will show that $\boldsymbol{E}_T[P(T)] = O(n \log n)$, for a randomly constructed binary search tree T with n nodes.

(a) *Show that the probability distribution on the set of binary search trees with n nodes that we obtain through random construction differs from the uniform distribution on this set.*
Hint*: Consider the case $n = 3$.*

(b) *Let $i, j \in \{1, \ldots, n\}$ with $i < j$. Let X_{ij} be the indicator random variable for the event that i is an ancestor of j or that j is an ancestor of i. Show that*

$$P(T) = \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} X_{ij}.$$

(c) *Let $i, j \in \{1, \ldots, n\}$ with $i < j$, and let X_{ij} as before. Show that*

$$\boldsymbol{E}[X_{ij}] = \frac{2}{j - i + 1}.$$

(d) *Show that*

$$\sum_{i=1}^{n-1} \sum_{j=i+1}^{n} \frac{2}{j - i + 1} = \sum_{i=2}^{n} \sum_{j=2}^{i} \frac{2}{j}.$$

(e) *Use the fact that $\sum_{j=2}^{i} \frac{1}{j} \leq \ln i$, for all $i \geq 2$, to get $\boldsymbol{E}_T[P(T)] = O(n \log n)$.*

2 Red-Black Trees and $(2,4)$-Trees (__/10)

A red-black tree *is a binary search tree in which every node has a color: red or black. The following rules apply: (i) the root is black; (ii) the $\perp$-nodes that indicate the empty subtrees are black; (iii) the children of every red node are black; (iv) the* black depth *of all $\perp$-nodes is the same, i.e., there is a number h such that for each $\perp$-node v, the number of black nodes on the path from v to the root is h.*

(a) *Show that a red-black tree with n nodes has height $O(\log n)$.*

(b) *Show that red-black trees and $(2,4)$-trees are equivalent. More precisely: describe a local transformation that translates groups of nodes in a red-black tree to nodes in a $(2,4)$-tree, and vice versa. Justify why your transformation fulfills the red-black tree properties and the $(2,4)$-tree properties.*

3 $(2,3)$-Trees (__/10)

(a) *Insert the keys A, L, P, D, R, E, I, X, Y, Z in this order into an initially empty $(2,3)$-tree. Show the individual steps. Afterwards, delete the keys A, L, X, R.*

(b) *Describe schematically a $(2,3)$-tree with n keys for which there is a deletion operation that requires $\Omega(\log n)$ rebalancing operations.*